

## AT32 PMSM FOC Sensorless Demo

## Introduction

This application note describes how to use a motor development board with vector control algorithm to drive a sensorless three-phase permanent magnet synchronous motor (brushless DC motor), covering project environment setup, program documentation structure, macro definition descriptions, and state machine workflow. ArteryTek offers a variety of motor development boards, and this document uses the AT-MOTOR-EVB motor development board paired with the AT32F413RCT7 sub-board as an example for explanation.

Applicable products:

|             |                      |
|-------------|----------------------|
| Part number | AT32F402/405 series  |
|             | AT32F403A/407 series |
|             | AT32F413 series      |
|             | AT32F415 series      |
|             | AT32F421 series      |
|             | AT32F422/426 series  |
|             | AT32F425 series      |
|             | AT32F423 series      |
|             | AT32F435/437 series  |
|             | AT32F45x series      |
|             | AT32L021 series      |
|             | AT32M412/M416 series |

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# 1 Motor control library algorithm

The motor control library algorithm mainly contains:

Target motor type: Three-phase permanent magnet synchronous motor (PMSM) / brushless DC motor (BLDC)

Control modes:

- FOC vector control

Three-phase PWM method:

- SVPWM
- Phase current sensing mode:
  - 3-shunt current sensing
  - 2-shunt current sensing
  - 1-shunt current sensing and reconstruction method

Rotor initial position estimation mode:

- Three-phase voltage vector mode

Estimation of rotor position in FOC driving mode:

- BEMF estimation with Luenberger observer

Sensorless FOC control mode:

- Open loop startup
- Torque control (current vector control)
- Speed control
- Field weakening control
- Upwind/Downwind startup

Auto tuning functions:

- Motor winding parameters identification
- Current PI controller parameter self-tuning

## 2 Environment requirements

### 2.1 Hardware requirement

A PMSM (BLDC) motor, AT-Link or third-party debugger and a motor control board (AT-MOTOR-EVB) are required. Refer to *AT32\_LV\_Motor\_Control\_EVB\_User\_Manual* (UM0011 for EVB\_V1.x or UM0014 for EVB\_V2.x) for relevant hardware configurations.

- PMSM (BLDC) motor: JK42BLS01-X056ED

Figure 1. JK42BLS01-X056ED motor



Table 1. Motor specifications

| Model                   | JK42BLS01-X056ED |
|-------------------------|------------------|
| Motor poles             | 8                |
| Rated voltage           | DC 24V           |
| Rated speed             | 4000 RPM         |
| Rated torque            | 0.0716 N·m       |
| No-load current         | 0.3 Amps         |
| Output power            | 30 W             |
| Line-to-line resistance | 1.8Ω             |
| No-load speed           | 6800 RPM         |
| Back EMF constant       | 4.36 V/Krpm      |
| Torque constant         | 0.042 N·m /A     |
| Encoder resolution      | 1000 P/R         |
| Insulation class        | Class B          |
| Winding type            | Delta            |

- Debugger
- Motor control board: AT-MOTOR-EVB V1.x or AT-MOTOR-EVB V2.x

Figure 2. AT-MOTOR-EVB\_V1.x evaluation board

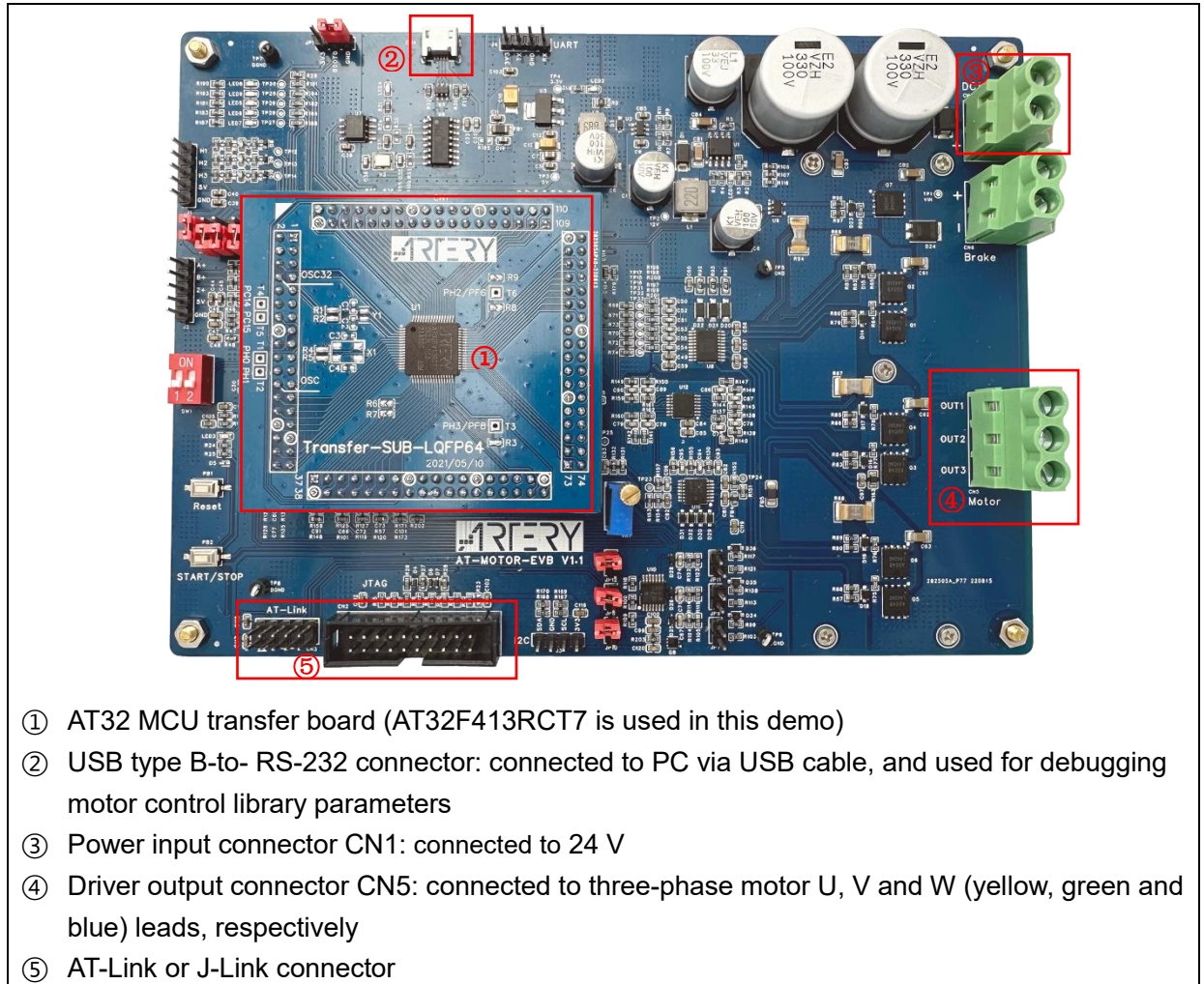
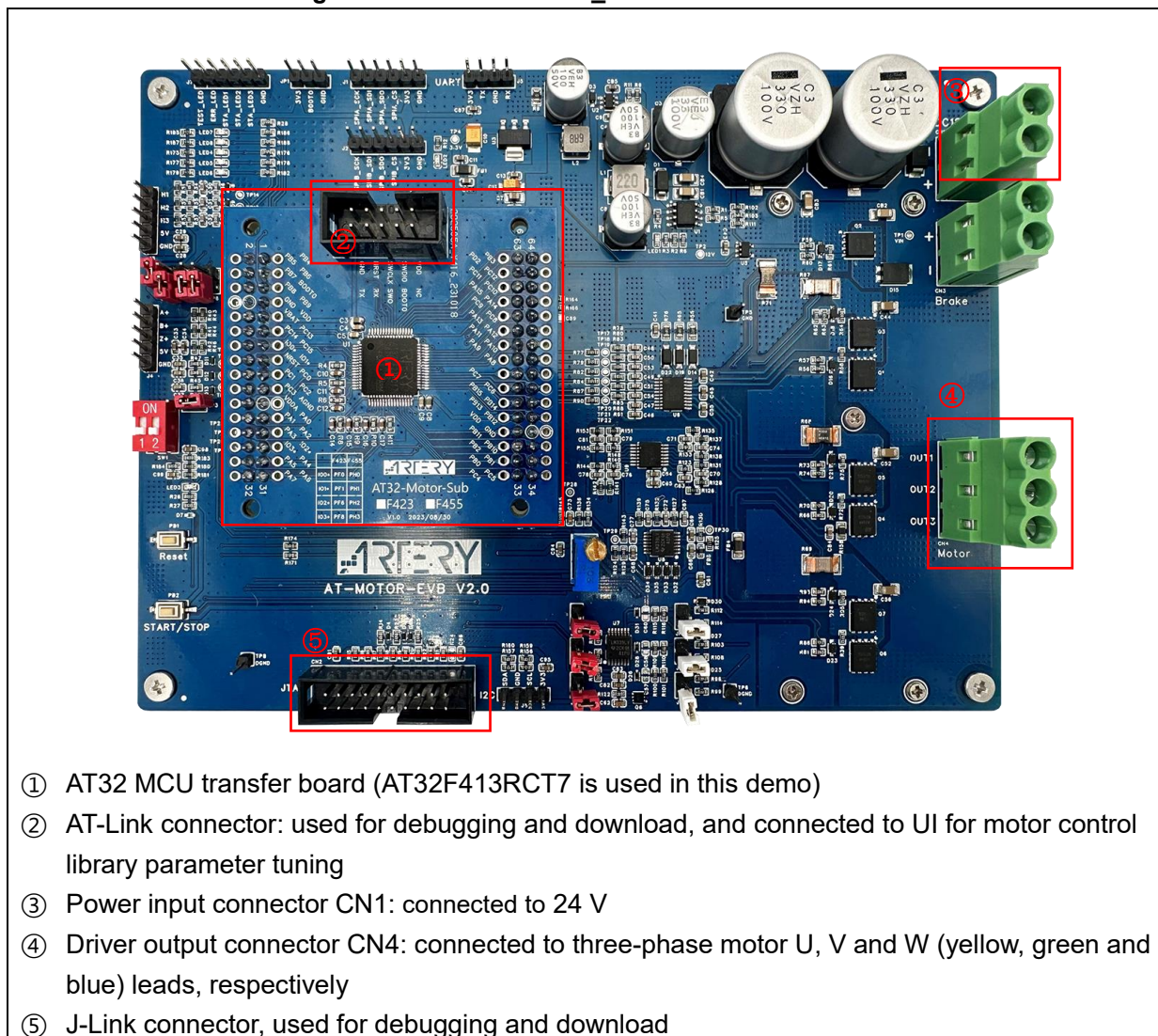


Figure 3. AT-MOTOR-EVB\_V2.x evaluation board



- ① AT32 MCU transfer board (AT32F413RCT7 is used in this demo)
- ② AT-Link connector: used for debugging and download, and connected to UI for motor control library parameter tuning
- ③ Power input connector CN1: connected to 24 V
- ④ Driver output connector CN4: connected to three-phase motor U, V and W (yellow, green and blue) leads, respectively
- ⑤ J-Link connector, used for debugging and download

## 2.2 Software requirement

- 1) Take the AT32F413 as an example. Open the demo project under AT32F413\_MC\_Library\_Project\motor\_evb\_2v0\at32f413\pmsm\_foc\_sensorless (Note[1]).
- 2) Run ArteryMotorMonitor.exe directly (installation not required).
- 3) In Keil IDE, go to **Options - Read/Only Memory Areas** to set ROM according to the Flash memory size of each AT32 MCU (see Table 2). For example, for AT32F413RCT7 with a 256KB Flash memory, its IROM1 start address is 0x8000000 with a size of 0x3F800, whereas its IROM2 start address is 0x803F800 with a size of 0x800 (see Figure 4). In AT32IDE, modify \*ld file according to the Flash memory size of each AT32 MCU (see Figure 5). In IAR Systems, modify \*icf file according to the Flash memory size of each AT32 MCU (see Figure 6).

**Table 2. Correspondence between Flash size and ROM**

| Flash size     | 4032K     | 1024K     | 512K      | 448K       | 256K      |
|----------------|-----------|-----------|-----------|------------|-----------|
| IROM1(address) | 0x8000000 | 0x8000000 | 0x8000000 | 0x8000000  | 0x8000000 |
| IROM1(size)    | 0x3EF000  | 0xFF800   | 0x7F800   | 0x6F000    | 0x3F800   |
| IROM2(address) | 0x83EF000 | 0x80FF800 | 0x807F800 | 0x0806F000 | 0x803F800 |
| IROM2(size)    | 0x1000    | 0x800     | 0x800     | 0x1000     | 0x800     |

| Flash size     | 128K      | 64K       | 32K       | 16K       |
|----------------|-----------|-----------|-----------|-----------|
| IROM1(address) | 0x8000000 | 0x8000000 | 0x8000000 | 0x8000000 |
| IROM1(size)    | 0x1FC00   | 0x0FC00   | 0x07C00   | 0x03C00   |
| IROM2(address) | 0x801FC00 | 0x800FC00 | 0x8007C00 | 0x8003C00 |
| IROM2(size)    | 0x400     | 0x400     | 0x400     | 0x400     |

*Note [1]: This demo project does not support Keil compiler version 5 and O0 optimization level for compiling simultaneously. Please use Keil compiler version 6 or O1 optimization level for compiling. In case of Keil\_v5.33, since AT32 BSP source code does not support V6.15 compiler, the Keil compiler version 5 and O1 optimization level should be used.*

Figure 4. AT32F413RCT7 ROM settings (Keil)

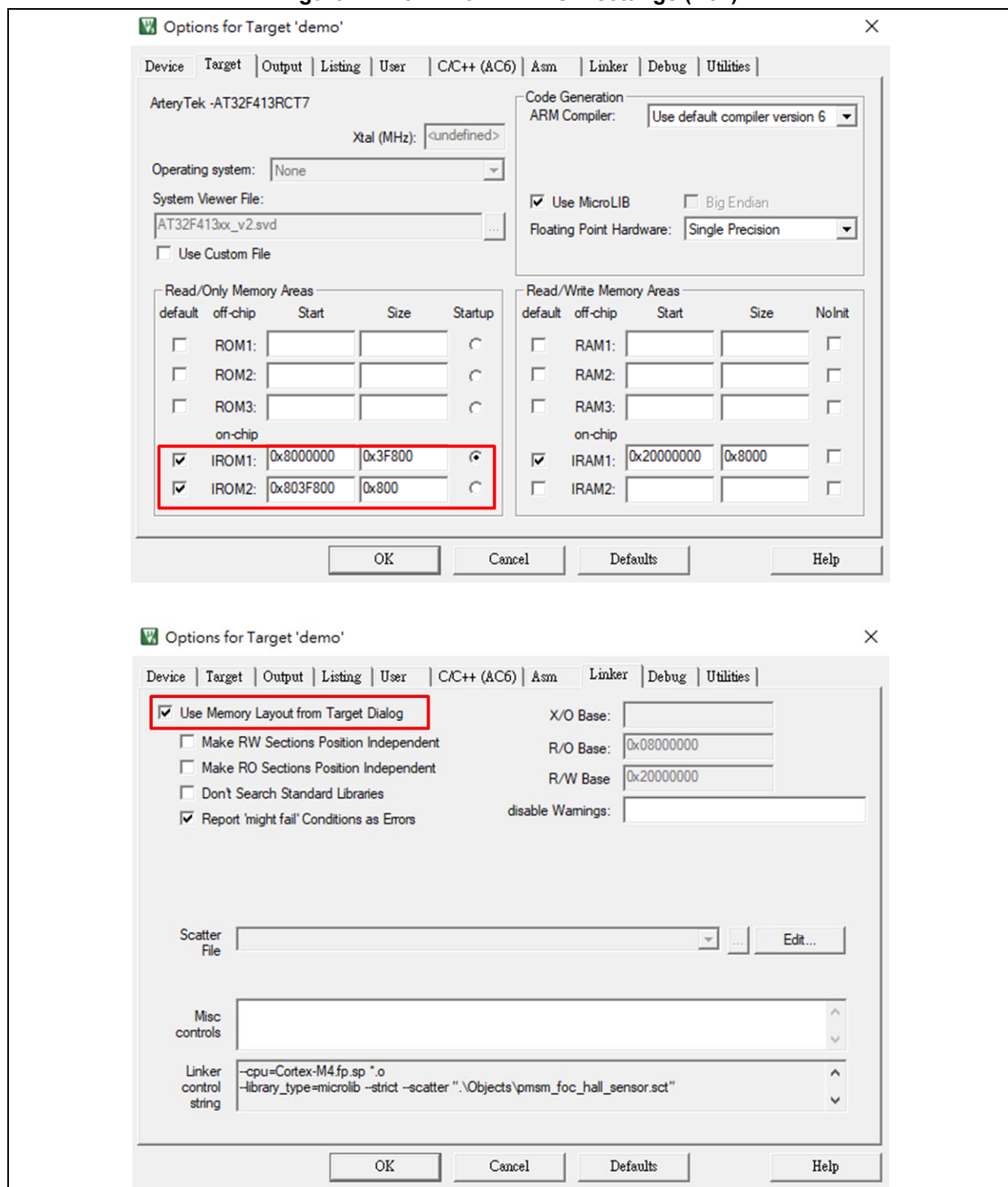


Figure 5. AT32F413RCT7 ROM setting (AT32IDE)

```

AT32F413xC_FLASH.ld X
25 _estack = 0x20008000; /* end of RAM */
26
27 /* Generate a link error if heap and stack don't fit into RAM */
28 _Min_Heap_Size = 0x200; /* required amount of heap */
29 _Min_Stack_Size = 0x400; /* required amount of stack */
30
31 /* Specify the memory areas */
32 MEMORY
33 {
34     FLASH (rx)      : ORIGIN = 0x08000000, LENGTH = 0x3F800
35     MC_DATA (r)      : ORIGIN = 0x0803F800, LENGTH = 0x800
36     RAM (xrw)       : ORIGIN = 0x20000000, LENGTH = 32K
37 }
38
39 /* Define output sections */
40 SECTIONS
41 {
42     /* The startup code goes first into FLASH */
43     .isr_vector :
44     {
45         . = ALIGN(4);
46         KEEP(*(.isr_vector)) /* Startup code */
47         . = ALIGN(4);
48     } >FLASH
49
50     .mc_data :
51     {
52         . = ALIGN(4);
53         . = ORIGIN(MC_DATA);
54         _MC_VectStoreAddr1 = .;
55         *(.MC_VectStoreAddr1);
56         . = ALIGN(4);
57         . = ORIGIN(MC_DATA) + 800;
58         _MC_VectStoreAddr2 = .;
59         *(.MC_VectStoreAddr2);
60         . = ALIGN(4);
61     } >MC_DATA
62
63     /* The program code and other data goes into FLASH */
64     .text :
65     {
66         . = ALIGN(4);
67         *(.text)           /* .text sections (code) */
68         *(.text*)          /* .text* sections (code) */
69         *(.glue_7)         /* glue arm to thumb code */
70         *(.glue_7t)        /* glue thumb to arm code */
71         *(.eh_frame)
72

```

Figure 6. AT32F413RCT7 ROM setting (IAR)

```

1  /****ICF**** Section handled by ICF editor, don't touch! ****/
2  /*-Editor annotation file-*/
3  /* IcfEditorFile="$TOOLKIT_DIR$\config\ide\IcfEditor\cortex_vl_0.xml" */
4  /*-Specials-*/
5  define symbol __ICFEDIT_intvec_start__ = 0x08000000;
6  /*-Memory Regions-*/
7  define symbol __ICFEDIT_region_ROM_start__ = 0x08000000;
8  define symbol __ICFEDIT_region_ROM_end__ = 0x0803F800 - 1;
9  define symbol __ICFEDIT_region_ROM1_start__ = 0x0803F800;
10 define symbol __ICFEDIT_region_ROM1_end__ = (__ICFEDIT_region_ROM1_start__ + 800 - 1);
11 define symbol __ICFEDIT_region_ROM2_start__ = (__ICFEDIT_region_ROM1_end__ + 1);
12 define symbol __ICFEDIT_region_ROM2_end__ = 0x0803F800 + 0x800 - 1;
13 define symbol __ICFEDIT_region_RAM_start__ = 0x20000000;
14 define symbol __ICFEDIT_region_RAM_end__ = 0x20007FFF;
15 /*-Sizes-*/
16 define symbol __ICFEDIT_size_cstack__ = 0x400;
17 define symbol __ICFEDIT_size_heap__ = 0x200;
18 /**** End of ICF editor section. ****ICF****/
19
20 define memory mem with size = 4G;
21 define region ROM_region = mem:[from __ICFEDIT_region_ROM_start__ to __ICFEDIT_region_ROM_end__];
22 define region ROM1_region = mem:[from __ICFEDIT_region_ROM1_start__ to __ICFEDIT_region_ROM1_end__];
23 define region ROM2_region = mem:[from __ICFEDIT_region_ROM2_start__ to __ICFEDIT_region_ROM2_end__];
24 define region RAM_region = mem:[from __ICFEDIT_region_RAM_start__ to __ICFEDIT_region_RAM_end__];
25
26 define block CSTACK with alignment = 8, size = __ICFEDIT_size_cstack__ { };
27 define block HEAP with alignment = 8, size = __ICFEDIT_size_heap__ { };
28
29 initialize by copy { readwrite };
30 do not initialize { section .noinit };
31
32 place at address mem:__ICFEDIT_intvec_start__ { readonly section .intvec };
33
34
35 place in ROM_region { readonly };
36 place in ROM1_region { readonly section .MC_VectStoreAddr1 };
37 place in ROM2_region { readonly section .MC_VectStoreAddr2 };
38 place in RAM_region { readwrite,
39 block CSTACK, block HEAP };

```

### 3 PMSM motor control library documents

Figure 7 shows the structure of motor control library documents and their relationships. The *motor\_control\_drive\_param.h* file contains the motor control drive modes, motor parameters, control board parameters and controller parameters to be set by users. Users can set MCU peripheral parameters in *mc\_hwio.h* file according to the connection relations between MCU peripherals and control board circuits. Relevant setting parameters are integrated in *mc\_foc\_globals.h* (*mc\_bldc\_globals.h*), and then their initial variables are set by functions in *mc\_foc\_globals.c* (*mc\_bldc\_globals.c*) and used by motor control library functions. For the MCU peripherals initialization, it is executed in *mc\_hwio.c*.

Figure 7. Structure of motor control library documents

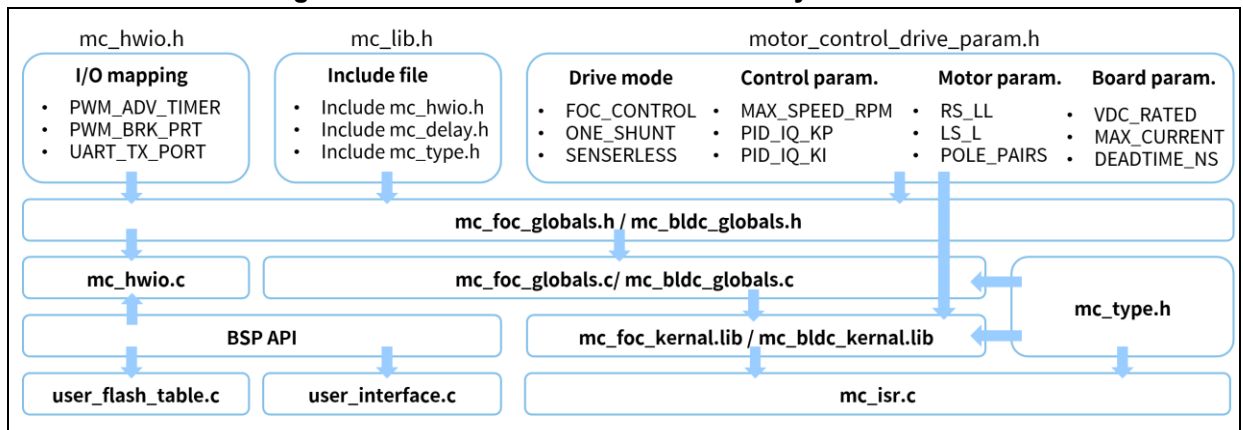


Figure 8 shows the document structure of the demo project. The “**user**” folder is user-defined and it contains the main program, peripheral program, writing parameters table into Flash, and parameter definition header files. The “**firmware**” folder contains BSP files, and the “**mcilib**” folder contains motor control library program, including delay functions, communication functions, global variable setting functions and motor control library functions. For details about motor control library functions, refer to *AN0064 AT32\_Motor\_Control\_Library\_User\_Guide*.

Figure 8. Structure of motor control project

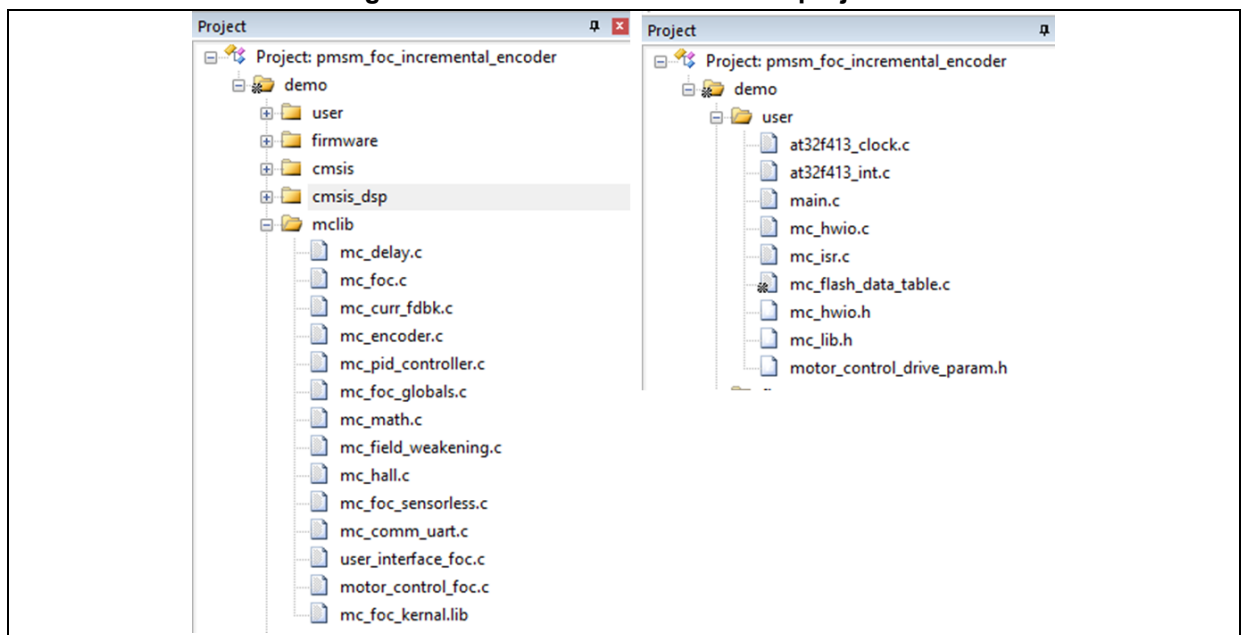


Table 3 lists motor control library documents.

**Table 3. Description of motor control library documents**

| File name                   | Description                                                                                                              |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| mc_lib.h                    | Unified header file management                                                                                           |
| motor_control_drive_param.h | Set motor drive modes (current sampling mode, sensor mode etc.), control parameters, driver parameters, motor parameters |
| mc_hwio.c                   | Hardware peripheral configuration                                                                                        |
| mc_hwio_v1.h                | Hardware IO macro definitions (AT-MOTOR-EVB V1.x)                                                                        |
| mc_hwio_v2.h                | Hardware IO macro definitions (AT-MOTOR-EVB V2.x)                                                                        |
| mc_isr.c                    | Relevant motor control interrupt functions                                                                               |
| mc_flash_data_table.c       | Writing parameters table into Flash                                                                                      |
| mc_flash_data_table.h       | Relevant configuration for writing parameters table into Flash                                                           |
| mc_type.h                   | Global variables types and definitions, enumeration definition                                                           |
| mc_delay.c                  | Delay-related functions                                                                                                  |
| mc_delay.h                  | Declaration of delay-related functions                                                                                   |
| mc_comm_uart.c              | Configuration of peripherals related to communication interface                                                          |
| mc_comm_uart.h              | Declaration and configuration of communication UART related functions                                                    |
| mc_pid_control.c            | PID controller related functions                                                                                         |
| mc_pid_control.h            | Declaration of PID controller related functions                                                                          |
| mc_curr_fdbk.c              | Current sensing related functions                                                                                        |
| mc_curr_fdbk.h              | Declaration of current sensing related functions                                                                         |
| mc_math.c                   | Filter related functions                                                                                                 |
| mc_math.h                   | Declaration of filter related functions                                                                                  |
| mc_hall.c                   | Hall sensor related functions                                                                                            |
| mc_hall.h                   | Declaration of Hall sensor related functions                                                                             |
| mc_foc_kernal.lib           | Motor control library core functions (for Keil environment only) (applicable to M4F MCU)                                 |
| mc_foc_kernal_noFPU.lib     | Motor control library core functions (for Keil environment only) (applicable to M4 MCU)                                  |
| mc_foc_kernal_m0plus.lib    | Motor control library core functions (for Keil environment only) (applicable to M0+ MCU)                                 |
| libmc_foc_kernal.a          | Motor control library core functions (for AT32IDE environment only) (applicable to M4F MCU)                              |
| libmc_foc_kernal_noFPU.a    | Motor control library core functions (for AT32IDE environment only) (applicable to M4 MCU)                               |
| libmc_foc_kernal_m0plus.a   | Motor control library core functions (for AT32IDE environment only) (applicable to M0+ MCU)                              |
| mc_foc_kernal.a             | Motor control library core functions (for IAR system only) (applicable to M4F MCU)                                       |

|                        |                                                                                    |
|------------------------|------------------------------------------------------------------------------------|
| mc_foc_kernal_noFPU.a  | Motor control library core functions (for IAR system only) (applicable to M4 MCU)  |
| mc_foc_kernal_m0plus.a | Motor control library core functions (for IAR system only) (applicable to M0+ MCU) |
| mc_foc_kernal.h        | Declaration of motor control library kernel functions                              |
| motor_control_foc.c    | Motor control related functions                                                    |
| motor_control_foc.h    | Declaration of motor control related functions                                     |
| mc_foc.c               | Vector control related functions                                                   |
| mc_foc.h               | Declaration of vector control related functions                                    |
| mc_encoder.c           | Encoder related functions                                                          |
| mc_encoder.h           | Declaration of encoder related functions                                           |
| mc_field_weakening.c   | Filed weakening related functions                                                  |
| mc_field_weakening.h   | Declaration of filed weakening related functions                                   |
| mc_foc_sensorless.c    | Sensorless mode related functions                                                  |
| mc_foc_sensorless.h    | Declaration of sensorless mode related functions                                   |
| mc_curr_decoupling.c   | Current decouple control related functions                                         |
| mc_curr_decoupling.h   | Declaration of current decouple control related functions                          |
| mc_foc_globals.c       | Global variables definition and default values, global functions definition        |
| mc_foc_globals.h       | Global variables, global functions definition, macro definition                    |
| user_interface_foc.c   | Functions related to communication interface                                       |
| user_interface_foc.h   | Declaration of functions related to communication interface                        |

1) *motor\_control\_drive\_param.h* file

- This file is divided into four parts, which are used to configure drive mode, control parameters, motor parameters and board parameters.
- Table 4 lists definitions of drive modes that can be set according to the hardware and motor configuration. In this demo, the sensorless FOC control mode is selected, and it has configurable open loop startup, align-and-go startup and initial angle detection startup. In sensorless FOC control mode, users can select whether or not to detect voltage input, and hardware filtering is required if voltage detection is selected (Note [2]). The current sampling mode (three-shunt, two-shunt or single-shunt current sampling mode) is set according to hardware configuration, and the field weakening control is optional.

**Table 4. Drive mode definitions**

| Definition item             | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| FOC_CONTROL                 | FOC vector control mode                                                |
| THREE_SHUNT                 | Three-shunt current sampling                                           |
| TWO_SHUNT                   | Two-shunt current sampling                                             |
| ONE_SHUNT                   | Single-shunt current sampling                                          |
| U_V_SHUNT                   | U&V shunt (for TWO_SHUNT only)                                         |
| V_W_SHUNT                   | V&W shunt (for TWO_SHUNT only)                                         |
| U_W_SHUNT                   | U&W shunt (for TWO_SHUNT only)                                         |
| AT_MOTOR_EVB_V1             | Motor control board V1.x (AT_MOTOR_EVB_V1.x)                           |
| AT_MOTOR_EVB_V2             | Motor control board V1.x (AT_MOTOR_EVB_V2.x)                           |
| GATE_DRIVER_LOW_SIDE_INVERT | Enable low side MOS inverted output                                    |
| FIELD_WEAKENING             | Field weakening control                                                |
| SENSORLESS                  | Sensorless control mode                                                |
| OPENLOOP_STARTUP            | Sensorless open loop startup                                           |
| ALIGN_AND_GO_STARTUP        | Align and go startup in sensorless control mode                        |
| INIT_ANGLE_STARTUP          | Initial angle detection startup in sensorless control mode             |
| VOLT_SENSE                  | Voltage sensing in sensorless vector control mode (Note [2])           |
| WIND_SENSE                  | Sensorless Upwind/Downwind Startup (Note [2])                          |
| CONST_CURRENT_START         | Constant current startup                                               |
| CONST_VOLTAGE_START         | Constant voltage startup                                               |
| CURRENT_LP_FILTER           | Get d/q axis current low-pass filter signals (vector control)          |
| AC_CURRENT_LP_FILTER        | Get A/B/C Phase Low-Pass Filtered Current Signals (Vector Control/FOC) |
| INTERNAL_CLOCK_SOURCE       | Use MCU internal oscillator as a clock source                          |
| MOTOR_PARAM_IDENTIFY        | Motor winding parameters identification                                |
| CURRENT_DECOUPLE_CTRL       | Current decouple control (NA for AT32L021 series)                      |
| BRAKING_RESISTOR            | Use an external braking resistor                                       |

|                    |                                                                                                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| SPMSM              | Surface-mounted permanent magnet synchronous motor                                                                                     |
| USE_MOTOR_MONITOR  | Artery motor control UI tool                                                                                                           |
| MOS_RDS_SHUNT      | Current Sampling Utilizing Low-Side MOSFET On-Resistance                                                                               |
| TWO_ADC_CONVERTERS | Current Sampling with Dual-ADC Synchronization<br>(Applicable for MCUs with dual or more ADCs; Not compatible with ONE_SHUNT topology) |

*Note [2]: Taking the AT-MOTOR-EVB V1.x development board as an example, components C67, C71, C74, C66, C70, and C72 must be mounted according to the schematic diagram. If the development board is AT-MOTOR-EVB V2.0, components C57, C58, and C60 should be mounted as the schematic. For the AT-MOTOR-EVB V2.1 development board, components C48, C49, and C50 must be mounted, and jumpers JP13, JP14, and JP15 need to be installed according to the schematic.*

- Table 5 lists the definitions of control parameters that can be defined according to the hardware, motor specifications and control characteristics. Some parameters are configurable, such as PI controllers of d-axis and q-axis currents, speed PI controller, etc.

**Table 5. Control parameter definitions**

| Definition item           | Description                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| PWM_FREQ                  | PWM output frequency (unit: Hz)                                                                                    |
| MOTOR_CONTROL_MODE        | Motor control mode (speed control, torque control, etc.; refer to motor_control_mode in <i>type.h</i> for details) |
| CTRL_SOURCE               | Control source settings (external source control/software control)                                                 |
| UI_UART_BAUDRATE          | Baud rate of UI communication serial port                                                                          |
| TUNE_TARGET_CURRENT       | Tune target current value for current loop PI control parameters (unit: ampere)                                    |
| TUNE_CURRENT_TOTAL_PERIOD | Tune total period of current pulse for current loop PI control parameters (unit: ms)                               |
| TUNE_CURRENT_STEP_PERIOD  | Tune current pulse step period for current loop PI control parameters (unit: ms)                                   |
| SPEED_LOOP_FREQ           | Speed control loop frequency (unit: Hz)                                                                            |
| MIN_SPEED_RPM             | Minimum motor speed (unit: rpm)                                                                                    |
| MAX_SPEED_RPM             | Maximum motor speed (unit: rpm)                                                                                    |
| ACC_SPD_SLOPE             | Slope of acceleration (unit: rpm/ms, systick =1kHz)                                                                |
| DEC_SPD_SLOPE             | Slope of deceleration (unit: rpm/ms, systick =1kHz)                                                                |
| SP_MAX_VOLT               | Maximum voltage of external command source (unit: voltage)                                                         |
| SP_THRESHOLD              | Threshold voltage of external command source (unit: voltage)                                                       |
| SP_RUN_VALUE              | Minimum voltage to startup in external source mode (unit: voltage)                                                 |
| SP_STOP_VALUE             | Maximum voltage to stop in external source mode (unit: voltage)                                                    |

| Definition item          | Description                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------|
|                          | voltage)                                                                                   |
| PID_SPD_KP_DIV           | Speed control proportional gain divisor (Q16 mode)                                         |
| PID_SPD_KI_DIV           | Speed control integral gain divisor (Q16 mode)                                             |
| PID_SPD_KP_DEFAULT       | Speed control proportional gain (Q15 mode)                                                 |
| PID_SPD_KI_DEFAULT       | Speed control integral gain (Q15 mode)                                                     |
| PID_ID_KP_DEFAULT        | d axis current control proportional gain (Q15 mode)                                        |
| PID_ID_KI_DEFAULT        | d axis current control integral gain (Q15 mode)                                            |
| PID_ID_KP_GAIN_DIV       | d axis current control proportional gain divisor (Q16 mode)                                |
| PID_ID_KI_GAIN_DIV       | d axis current control integral gain divisor (Q16 mode)                                    |
| PID_IQ_KP_DEFAULT        | q axis current control proportional gain (Q15 mode)                                        |
| PID_IQ_KI_DEFAULT        | q axis current control integral gain (Q15 mode)                                            |
| PID_IQ_KP_GAIN_DIV       | q axis current control proportional gain divisor (Q16 mode)                                |
| PID_IQ_KI_GAIN_DIV       | q axis current control integral gain divisor (Q16 mode)                                    |
| OLC_ANGLE_INC            | Open loop control angle incremental at each PWM output frequency                           |
| OLC_VOLT                 | Open loop control voltage output (Q15 mode)                                                |
| FW_MAX_ID_CURR           | Maximum d-axis current in field weakening control (unit: ampere)                           |
| FW_KP_GAIN               | Field weakening control proportional gain (Q15 mode)                                       |
| FW_KI_GAIN               | Field weakening control integral gain (Q15 mode)                                           |
| FW_KP_GAIN_DIV           | Field weakening control proportional gain divisor (Q16 mode)                               |
| FW_KI_GAIN_DIV           | Field weakening control integral gain divisor (Q16 mode)                                   |
| AUTO_TUNE_CURR_BANDWIDTH | Current PI controller bandwidth (used for self-tuning of current PI controller parameters) |
| CURR_LP_BANDWIDTH        | d/q-axis current low-pass filter bandwidth (unit: Hz)                                      |
| AC_CURR_LP_BANDWIDTH     | a/b/c phase current low-pass filter band width (unit: Hz)                                  |
| OBS_SPD_LP_BANDWIDTH     | Observer speed low-pass filter bandwidth (unit: Hz)                                        |
| OBS_GAIN1                | Observer gain factor 1 (Q15 mode)                                                          |
| OBS_GAIN2                | Observer gain factor 2 (Q15 mode)                                                          |
| PLL_KP_GAIN              | PLL control proportional gain (Q15 mode)                                                   |
| PLL_KI_GAIN              | PLL control integral gain (Q15 mode)                                                       |
| PLL_KP_GAIN_DIV          | PLL control proportional gain divisor (Q16 mode)                                           |
| PLL_KI_GAIN_DIV          | PLL control integral gain divisor (Q16 mode)                                               |
| STARTUP_MAX_SPD          | Open loop maximum speed before entering to closed loop (unit: rpm)                         |
| STARTUP_CURRENT          | Current command before entering closed loop (unit: ampere)                                 |
| STARTUP_VOLTAGE          | Voltage command before entering closed loop (unit: voltage)                                |

| Definition item      | Description                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------|
| STARTUP_OL_SLOPE     | Open loop startup acceleration (unit: rpm/s)                                                     |
| STARTUP_ALIGN_TIME   | Rotor alignment time before startup (unit: ms)                                                   |
| STARTUP_START_TIME   | Startup time (switching to closed loop) after rotor alignment (unit: ms)                         |
| DETECT_PULSE_WIDTH   | Voltage pulse width for rotor initial angle detection (unit: us)                                 |
| DETECT_DELAY_TIME    | Sensorless startup detection of delay time in motor reversal (unit: ms)                          |
| DETECT_REVERSE_SPEED | Detect motor reverse speed in sensorless startup mode (unit: rpm)                                |
| DETECT_MAX_SPEED     | Detect the motor maximum forward speed in sensorless startup mode (unit: rpm)                    |
| WIND_DETECT_TIME     | Upwind/Downwind detection time (unit: ms)                                                        |
| TAILWIND_SPEED       | Minimum speed for downwind closed-loop engagement (unit: rpm)                                    |
| HEADWIND_BRAKE_TIME  | Upwind braking time (unit: ms)                                                                   |
| HIGH_PWM_DUTY_CYCLE  | Dynamic Adjustment of Current Sampling Position at High PWM Duty Cycle (TWO_ADC_CONVERTERS only) |
| MAX_DUTY_PCT         | Maximum PWM Duty Cycle Setting                                                                   |

- Table 6 lists drive-related parameter definitions, such as dead time, shunt resistor and current amplifier gain. The corresponding parameters should be modified if other motor drive board is used.

Table 6. Driver parameter definitions

| Definition item          | Description                                                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| VDC_RATED                | DC-BUS voltage (unit: voltage)                                                                                                         |
| V_SENSE_GAIN             | Divider ratio of voltage feedback                                                                                                      |
| ADC_REFERENCE_VOLT       | ADC reference voltage (unit: voltage)                                                                                                  |
| ADC_DIGITAL_SCALE_12BITS | ADC resolution                                                                                                                         |
| SYSTEM_CORE_CLOCK        | System frequency (unit: Hz)                                                                                                            |
| TMR_CLK                  | Timer clock frequency (unit: Hz)                                                                                                       |
| DEADTIME_CLK_SFT_BITS    | Shift bit number of frequency divider for dead time generator                                                                          |
| DEADTIME_NS              | Dead time (unit: ns)                                                                                                                   |
| MIN_INTERVAL_TIME        | Minimum interval time between two-phase signals during PWM shift (unit: ns)                                                            |
| ADC_TRIG_DELAY_TIME      | Current sampling delay time (unit: us)                                                                                                 |
| ADC_TRIG_LAG_TIME        | ADC trigger lag time for current sensing (Two-Phase Low-Level Condition) (Unit: ns) (only for TWO_ADC_CONVERTERS && THREE_SHUNT)       |
| SW_OP_INP_MODE_LEAD_TIME | OP Input Mode Switching Lead Time Prior to ADC Trigger (for chips with internal OP like M412)                                          |
| MAX_CURRENT              | Maximum peak phase current of motor (unit: ampere)                                                                                     |
| MIN_CURRENT              | Minimum peak phase current of motor (unit: ampere)                                                                                     |
| CURRENT_SPAN_SHIFT       | Number of shifts for current per-unit normalization                                                                                    |
| R_SHUNT                  | Shunt resistance (unit: $\Omega$ )                                                                                                     |
| OP_GAIN                  | Current amplifier gain                                                                                                                 |
| CURR_OFFSET_VOLT         | Zero current offset (unit: voltage)                                                                                                    |
| EMF_SENSE_GAIN           | Divider ratio of BEMF feedback                                                                                                         |
| OVER_CURRENT_SW          | Software Overcurrent Threshold (unit: ampere)                                                                                          |
| DAC_VREF_SOURCE          | DAC Source Selection for Internal OP Overcurrent Protection (Dedicated to chips with internal OP, e.g., M412 series)                   |
| OCP_CURRENT              | Hardware Overcurrent Threshold (unit: ampere)                                                                                          |
| BUS_CURR_CMP_OCP_VOLT    | Internal OP Overcurrent Protection Voltage (unit: volt)<br>(Dedicated to chips with internal operational amplifier, e.g., M412 series) |
| TMR_BRK_FILTER_COUNT     | Overcurrent Protection Break Input Filter Count<br>(Dedicated to chips with break input function, e.g., M412 series)                   |
| OVER_VOLT_THRESHOLD      | Over-voltage threshold point (unit: voltage)                                                                                           |
| UNDER_VOLT_THRESHOLD     | Under-voltage threshold point (unit: voltage)                                                                                          |

| Definition item     | Description                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| V0_V                | Parameter V0 of the temperature and voltage linear approximation function of NTC voltage divider circuit (Note [3])    |
| T0_C                | Parameter T0 of the temperature and voltage linear approximation function of NTC voltage divider circuit (Note [3])    |
| dV_dT               | Parameter dV/dT of the temperature and voltage linear approximation function of NTC voltage divider circuit (Note [3]) |
| OVER_TEMP_THRESHOLD | Over-temperature threshold point (unit: Celsius degrees)                                                               |
| MC_ERROR_MASK       | Error detection mask                                                                                                   |

Note [3]: Approximate curve equation of voltage-temperature relation is  $V[V]=V0+dV/dT[V/Celsius]*(T-T0)[Celsius]$ .

- Table 7 lists definitions of motor parameters, including number of pole-pairs, encoder parameters and Hall sensor parameters.

Table 7. Motor parameter definitions

| Definition item | Description                                             |
|-----------------|---------------------------------------------------------|
| RS_LL           | Motor line-to-line winding resistance (unit: $\Omega$ ) |
| LS_LL           | Motor line-to-line winding inductance (unit: mH)        |
| KE              | Back EMF constant (KE) (unit: V/rpm)                    |
| LD_LQ_RATIO     | Ld to Lq ratio                                          |
| POLE_PAIRS      | Number of pole-pairs                                    |
| NOMINAL_CURRENT | Motor nominal current (unit: ampere)                    |

## 2) *mc\_hwio.h* file

- This file is mainly used for configuring definitions based on the mapping of user hardware IO interface and MCU peripherals, and it also includes function declarations in *mc\_hwio.c* file. Table 8 lists the configurations of peripherals such as ADC, TMR, USART and EXINT, interrupt functions and the corresponding DMA channels.

Table 8. Configuration of peripherals (AT32F413RCT7)

| Peripheral | Interrupt function            | DMA channel | Description                                                                                                                              |
|------------|-------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------|
| ADC1       | N/A                           | DMA1_CH1    | ADC1 ordinary channel conversion (BUS voltage, MOS temperature, external speed/torque command, three-phase BEMF voltage and BUS current) |
| ADC1       | ADC1_2_IRQn                   | N/A         | ADC1 preempted channel conversion (phase current or bus current, and initial angle detection current)                                    |
| TMR1       | TMR1_OVF_TMR10_IRQn           | N/A         | FOC control interrupt                                                                                                                    |
| TMR1       | TMR1_BRK_TMR9_IRQn            | N/A         | PWM output disable                                                                                                                       |
| TMR1_CH4   | N/A                           | DMA1_CH4    | Set sampling time point configuration in one-shunt mode                                                                                  |
| TMR11      | TMR1_TRG_HALL_TMR11_IRQHandle | N/A         | Speed control loop interrupt                                                                                                             |
| USART1_TX  | N/A                           | DMA1_CH2    | USART1 TX data                                                                                                                           |
| USART1_RX  | N/A                           | DMA1_CH3    | USART1 RX data                                                                                                                           |
| EXINT      | EXINT15_10_IRQn               | N/A         | START/STOP button (start/stop motor)                                                                                                     |

## 3) *mc\_hwio.c* file

- This file is mainly configured the MCU peripherals based on user hardware, such as TMR, ADC, DMA and GPIO. It also includes some basic control functions such as button, LED, DIP switch. Refer to Table 9 for details.

**Table 9. Peripheral configuration related functions**

| Function                               | Description                                                                                    |
|----------------------------------------|------------------------------------------------------------------------------------------------|
| <code>nvic_config</code>               | Interrupt priority configuration                                                               |
| <code>tmr_pwm_init</code>              | PWM timer, crm clock, GPIO and DMA configuration                                               |
| <code>adc_ordinary_config</code>       | ADC ordinary channel ADC, DMA and GPIO configuration                                           |
| <code>adc_preempt_config</code>        | ADC preempted channel ADC, DMA and GPIO configuration                                          |
| <code>speed_timer_init</code>          | Speed control timer and crm clock configuration                                                |
| <code>uart_init</code>                 | UART crm clock, GPIO and UART configuration                                                    |
| <code>button_exint_init</code>         | Button interrupt event EXINT configuration                                                     |
| <code>led_config</code>                | LED GPIO initialization and crm clock configuration                                            |
| <code>led_on</code>                    | LED ON                                                                                         |
| <code>led_off</code>                   | LED OFF                                                                                        |
| <code>led_toggle</code>                | LED toggle (from ON to OFF, or from OFF to ON)                                                 |
| <code>led_init</code>                  | LED initialization configuration                                                               |
| <code>led_blink</code>                 | LED blinks                                                                                     |
| <code>mode_switch_init</code>          | DIP switch configuration                                                                       |
| <code>foc_angle_init_config</code>     | Initial angle detection related timer and ADC configuration                                    |
| <code>motor_parameter_ID_config</code> | Configure TMR and ADC relevant to motor winding parameter identification                       |
| <code>hwdiv_config</code>              | Hardware divider configuration (for AT32L021 only)                                             |
| <code>get_int_vref_cal_ratio</code>    | Detect VDDA reference voltage ratio, used for ADC value calibration                            |
| <code>opa_init</code>                  | Busbar Current Amplification OP Configuration (AT32M412 and other chips with internal OP only) |
| <code>opa_calibration</code>           | Busbar Current Amplification OP Calibration (AT32M412 and other chips with internal OP only)   |
| <code>ocp_cmp_config</code>            | Busbar Current Protection CMP Configuration (AT32M412 and other chips with internal CMP only)  |

4) *mc\_isr.c* file

- This file includes interrupt handler functions for ADC, TMR and SYSTICK interrupts, etc. Refer to Table 10.

**Table 10. Motor control related interrupt functions**

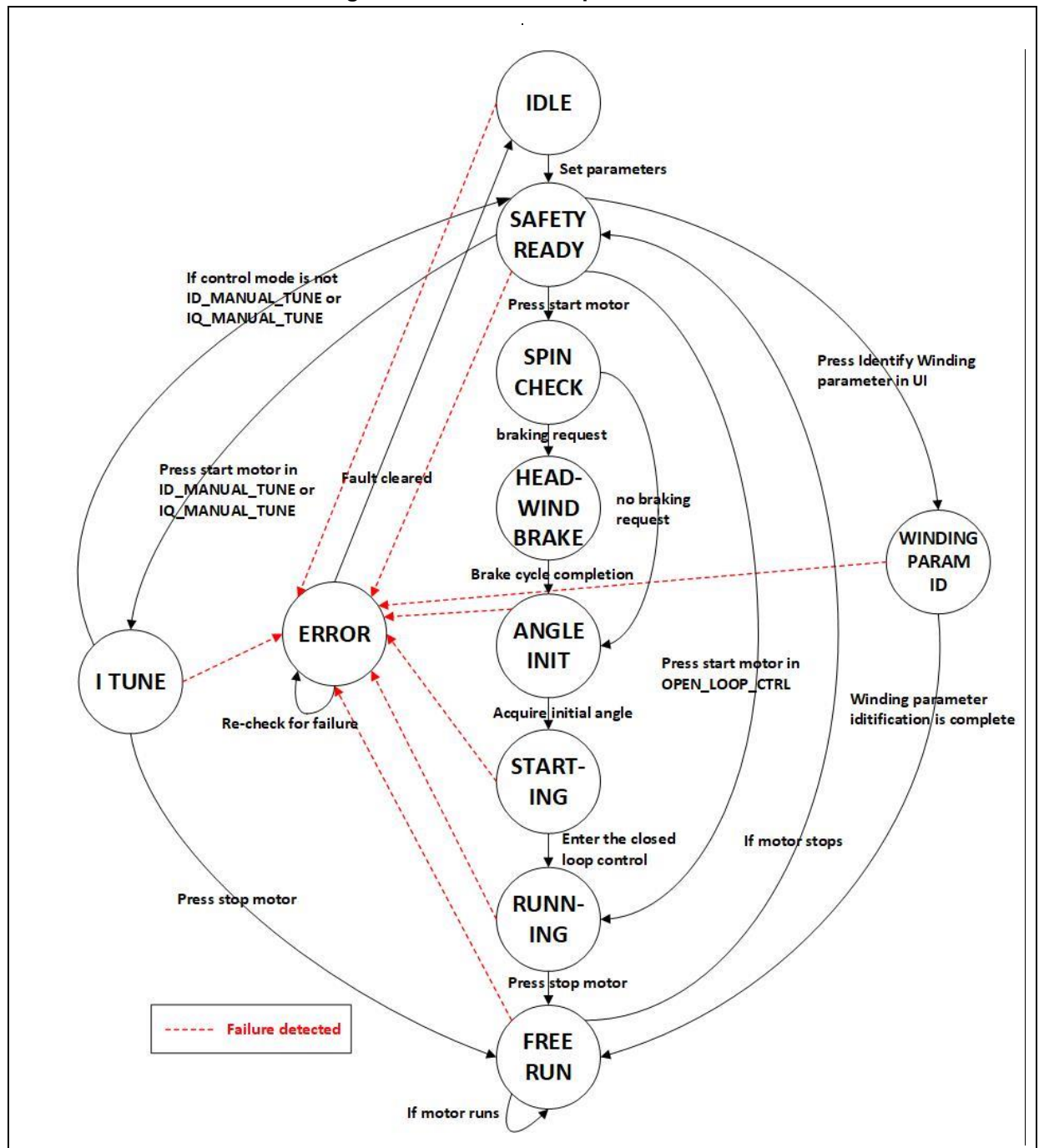
| Function                                                 | Description                                                                                                          |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| ADVTMR_PWM_CYCLE_IRQ<br>(TMR1_OVF_TMR10_IRQHandler)      | TIMER1 update interrupt function: speed and angle estimation, vector control open loop control, current loop control |
| ADVTMR_PWM_BRK_IRQ<br>(TMR1_BRK_TMR9_IRQHandler)         | TMR1 break input interrupt function: PWM output disable                                                              |
| ADC_SHUNT_SAMP_READY_IRQ<br>(ADC1_2_IRQHandler)          | ADC1 interrupt function: current sampling complete                                                                   |
| SPEED_LOOP_TIMER_IRQ<br>(TMR1_TRG_HALL_TMR11_IRQHandler) | Speed control loop interrupt function                                                                                |
| SysTick_Handler                                          | System interrupt function (1ms): includes state machine program                                                      |
| BUTTON_EXINT_IRQ<br>(EXINT15_10_IRQHandler)              | External IO interrupt function: START/STOP button (start/stop motor)                                                 |

## 4 State machine

### 4.1 Description of state

Figure 9 shows the state machine of the application example of motor control library. The initial settings of each state are constantly checked by the Main program; when the state changes, the initial settings are set for the new state. In addition, the state machine runs in the SysTick Handler interrupt function every minisecond to ensure that it is checked in real time. Each state is checked for fault; once detected, the state machine enters “Error” state and stop driving to avoid damage to the motor or motor drive board, and the motor stops until the fault is cleared.

Figure 9. State machine process flow



It includes Idle, Safety ready, Angle init, Starting, Running, Free run, I\_tune, Error, Spin check and

Headwind brake as detailed below.

**Idle**

It is the initial state of state machine, and motor idles in this state. The state machine returns to "Idle" state when the error condition is released.

**Safety ready**

This state indicates that the motor can be started up safely after all parameters are set and checked and the current offset is obtained in the "Idle" state.

**Angle init**

It is the exclusive state of the sensorless control mode (INIT\_ANGLE\_STARTUP). The program enters this state for detecting the initial angle before starting the motor, and then jumps to the "Starting" state if there is no INIT\_ANGLE\_STARTUP.

**Starting**

In the sensorless control mode, enter this state to set the operation mode parameters, and peripheral configuration. In addition, set the conditions for switching to closed-loop control to improve system stability.

**Running**

The motor is running in this state. Through the UI software, users can on-line adjust parameters (target speed, target current, etc.) or stop the motor.

**Free run**

In this state, the drive stops voltage output, and the motor speed will slowly down to zero. Program remains in this state until the motor stops completely, and then returns to the "Safety ready" state.

**I\_tune**

This state is used for adjusting PID controller parameters. In this state, users can adjust the target current, current loop KP and KI values through UI software. A step current is generated in this state, based on which parameters are adjusted.

**Winding param Id**

This refers to the state of motor winding parameter identification. Users can enable this feature via UI interface. The winding parameter identification feature is often used for tuning a new motor. The tuned parameters can be used for sensorless vector control and current PI controller parameters self-tuning.

**Error**

Jump to this state in case of any error.

**Spin check (Note[4])**

This mechanism is designed for rotation direction identification and speed monitoring during the motor startup phase. When the WIND\_SENSE macro (enabling headwind/tailwind detection) is activated, the system prioritizes executing a pre-start self-test procedure. The estimator first evaluates the motor's rotational speed and direction, then determines whether to initiate motor

braking or directly engage closed-loop startup.

**Headwind brake (Note[4])**

During the motor startup phase, the system prioritizes Spin Check State to monitor the motor's rotational speed and direction. If both of the following conditions are met:

1. Motor rotation direction is headwind (counter to wind flow)
2. Rotational speed is below the negative closed-loop engagement threshold (-TAILWIND\_SPEED macro)

The system enters this state to execute three-phase lower MOSFET arm short-circuit braking. Upon completing the predefined braking duration (HEADWIND\_BRAKE\_TIME macro), it transitions to Starting State or Angle Init State (the transition path is determined by the user-configured startup mode macro)

*Note[4] :When the WIND\_SENSE macro (headwind/tailwind detection) is enabled, the system prioritizes the pre-start self-test sequence (Spin Check state). If braking is required, it transitions to the Headwind Brake State for braking the motor.*

Table 11 lists each state in details and state switching conditions.

**Table 11. Initial state and state switching conditions**

| Initial state | End state        | State switching conditions                                                                                                                                                                                                                                                                           |
|---------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDLE          | SAFETY REDAY     | The control source is the UI tool, and initial parameters are set.                                                                                                                                                                                                                                   |
|               |                  | The control source is the motor control board, POTENTIOMETER command is given less than the default, and initial parameters are set.                                                                                                                                                                 |
| SAFETY REDAY  | SPIN CHECK       | When the control mode is not ID_MANUAL_TUNE, IQ_MANUAL_TUNE or OPEN_LOOP_CTRL, click START MOTOR (UI tool) or START/STOP button on the motor control board.                                                                                                                                          |
|               | ANGLE INIT       | When ALL of the following are true:<br>1. Control mode $\neq$ ID_MANUAL_TUNE, IQ_MANUAL_TUNE, or OPEN_LOOP_CTRL<br>2. WIND_SENSE macro disabled<br>3. INIT_ANGLE_STARTUP macro enabled<br>Activating either:<br>1. START MOTOR button on UI tool<br>2. START/STOP button on the motor control board  |
|               | STARTING         | When ALL of the following are true:<br>1. Control mode $\neq$ ID_MANUAL_TUNE, IQ_MANUAL_TUNE, or OPEN_LOOP_CTRL<br>2. WIND_SENSE macro disabled<br>3. INIT_ANGLE_STARTUP macro disabled<br>Activating either:<br>1. START MOTOR button on UI tool<br>2. START/STOP button on the motor control board |
|               | RUNNING          | When the control mode is OPEN_LOOP_CTRL, click START MOTOR (UI tool) or START/STOP button on the motor control board.                                                                                                                                                                                |
|               | I TUNE           | When the control mode is ID_MANUAL_TUNE or IQ_MANUAL_TUNE, click START MOTOR (UI tool) or START/STOP button on the motor control board.                                                                                                                                                              |
|               | WINDING_PARAM_ID | Click Identify Winding parameter (UI tool)                                                                                                                                                                                                                                                           |
| SPIN CHECK    | HANDWIND BRAKE   | When BOTH of the following are true:<br>1. Motor rotation direction is upwind (counter-wind)<br>2. Rotational speed < -TAILWIND_SPEED (negative value indicates reverse direction)                                                                                                                   |
|               | ANGLE INIT       | When the INIT_ANGLE_STARTUP macro is enabled, the motor rotation direction is either downwind or upwind, and the                                                                                                                                                                                     |

|                  |              |                                                                                                                                                                                |
|------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  |              | rotational speed is within the range of $\pm$ TAILWIND_SPEED                                                                                                                   |
|                  | STARTING     | When the INIT_ANGLE_STARTUP macro is disabled, the motor rotation direction is either downwind or upwind, and the rotational speed is within the range of $\pm$ TAILWIND_SPEED |
|                  | RUNNING      | the motor rotation direction is downwind and the rotational speed $\geq$ TAILWIND_SPEED                                                                                        |
| HANDWIND BRAKE   | ANGLE INIT   | When the INIT_ANGLE_STARTUP macro is enabled and braking time exceeds HEADWIND_BRAKE_TIME                                                                                      |
|                  | STARTING     | When the INIT_ANGLE_STARTUP macro is disabled and braking time exceeds HEADWIND_BRAKE_TIME                                                                                     |
| ANGLE INIT       | STARTING     | Complete initial angle detection.                                                                                                                                              |
| STARTING         | RUNNING      | Open loop control or alignment startup ends and enter closed loop control.                                                                                                     |
| RUNNING          | FREE_RUN     | Click STOP MOTOR (UI tool) or START/STOP button on the motor control board.                                                                                                    |
| FREE_RUN         | SAFETY REDAY | Motor stops.                                                                                                                                                                   |
|                  | FREE_RUN     | Motor runs.                                                                                                                                                                    |
| I TUNE           | SAFETY REDAY | Control mode is not ID_MANUAL_TUNE nor IQ_MANUAL_TUNE.                                                                                                                         |
|                  | FREE_RUN     | Click STOP MOTOR (UI tool) or START/STOP button on the motor control board.                                                                                                    |
| WINDING_PARAM_ID | FREE_RUN     | Motor winding parameter identification is completed.                                                                                                                           |
| ERROR            | IDLE         | Fault is cleared.                                                                                                                                                              |

## 5 How to use demo project

For details, refer to *AN0220 AT32\_PMSM\_FOC\_Sensorless\_Quick\_Guide*.

## 6 Revision history

Table 12. Document revision history

| Date       | Version | Revision note                                                                                                                                                                                                                                                               |
|------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023.01.13 | 2.0.0   | Initial release.                                                                                                                                                                                                                                                            |
| 2023.04.20 | 2.0.1   | Updated operation instruction of the latest version of UI program.                                                                                                                                                                                                          |
| 2023.10.03 | 2.0.2   | Updated description of motor control library documents, and added applicable products in the first page.                                                                                                                                                                    |
| 2024.02.27 | 2.0.3   | Added AT-MOTOR-EVB V2.0 wiring descriptions, macro definitions related to auto identification of motor winding parameters and self-tuning of current PI controller parameters, updated description of state machine, and added instructions to motor parameters adjustment. |
| 2024.05.24 | 2.1.0   | Updated tuning procedures and descriptions.                                                                                                                                                                                                                                 |
| 2024.12.27 | 2.1.1   | Added applicable products, including AT32F435/437 and AT32L021 series;<br>Added IAR system and the corresponding Flash ROM configuration;<br>Updated some definitions and description.                                                                                      |
| 2025.04.11 | 2.1.2   | Added applicable products: AT32F425 series;<br>Added macro definitions and state machine descriptions for sensorless upwind/downwind startup                                                                                                                                |
| 2025.10.09 | 2.1.3   | Added constant voltage startup mode, added supported models, corrected flash configuration table, and updated parameters.                                                                                                                                                   |

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